

CELLGUARD · DESIGN DECISION RECORD

CellGuard TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellguard		
Source	docs/decisions/0001-trl2-review-decisions.md @ c58dc26		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted for TRL 3 under Amish's 2026-09-25 instruction, open for his review, and the items that stay open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. The recommendations in items 1 to 8 are adopted for TRL 3 work pending Amish's review; items 9 to 15 remain proposed, awaiting Amish.

■ Context

The TRL 2 review note (`docs/REVIEW.md` , session of 2026-09-25) listed nine items as "Proposed, awaiting Amish", eight of them with a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed the CellGuard items one by one. Under that instruction, each item that has a recommendation is adopted as recommended for TRL 3, open for his review; items without a recommendation stay open. Budgets are not changed in `project.yaml` : a recommended budget is recorded here as awaiting Amish. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in `docs/REVIEW.md` (TRL 2 session) and CGD-PRC-001 v0.2. They are not repeated here.

■ Decision

Table 1. Items adopted for TRL 3 work.

#	Item	Status and content	Where it now lives
1	Front end	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. TI BQ76952 class (3 to 16 cells, autonomous protection, integrated high-side driver and coulomb counter)	CGD-PRC-001 v0.3, <code>bom/bom.csv</code> line 3, CGD-CAL-001
2	Microcontroller and firmware base	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. Option A: STM32G0B1 class with CAN, building on the Libre Solar BMS firmware (Apache-2.0). How Apache-2.0 code sits in this MIT-licensed repo stays open (item 10)	CGD-PRC-001 v0.3, <code>bom/bom.csv</code> line 6
3	Secondary protector (safety trade-off)	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. Add a BQ77216-class independent protector (overvoltage, undervoltage, open wire, temperature) driving a self-control protector (SCP) fuse in the B+ path, about \$10	CGD-PRC-001 v0.3, <code>bom/bom.csv</code> line 14, <code>cad/src/model.py</code> part 14, CGD-CAL-001 section 4
4	Budget	Recommendation (a), raise <code>budget_usd</code> to \$140, is Proposed, awaiting Amish . <code>budget_usd</code> stays at \$120 in <code>project.yaml</code> . CGD-CAL-001 states the cost against both figures: \$134 is 11.7 % over \$120 and \$6 under \$140	CGD-REQ-001 R16, CGD-CAL-001 section 11
5	Current rating	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. 40 A continuous for the first board; a 100 A variant later only if storage users need it	CGD-REQ-001 R5 (unchanged)
6	Switching side	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. High-side N-channel switching, keeping a common ground for CAN	CGD-PRC-001 v0.3
7	Chemistry scope	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. Hardware designed for LFP and for an NMC profile (up to 14S under 60 V, which covers SwapCell's 13S), LFP firmware released first. Use as SwapCell's BMS still needs the SwapCell project's agreement. The pitch is unchanged	CGD-REQ-001 R1, CGD-PRB-001 v0.3, CGD-PRC-001 v0.3
8	CAN message set	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. CellGuard carries the SwapCell interface v0.3 message set (250 kbit/s, 11-bit identifiers) as a profile, subject to the SwapCell project	CGD-REQ-001 R11, CGD-PRC-001 v0.3, CGD-CAL-001 section 8

No pitch or problem rewording was recommended at TRL 2, so `project.yaml` and the README keep their pitch and problem lines.

Items that remain open

Table 2. Items still Proposed, awaiting Amish.

#	Item	Status
9	First test partner (solar installer, e-bike repair shop or makerspace)	Proposed, awaiting Amish. No preference stated
10	Licensing of Apache-2.0 firmware code in an MIT-licensed repo	Proposed, awaiting Amish. Suggested for his review: keep reused files under Apache-2.0 with their notices in a separate firmware folder and list them in <code>LICENSE-SOFTWARE</code> . No choice has been made
11	Budget figure (item 4)	Proposed, awaiting Amish: \$140 recommended
12	Cover material: printed flame-retardant polycarbonate or sheet aluminium	Proposed, awaiting Amish. No recommendation was made
13	Low-cost method to check the state-of-charge estimate (for example a calibrated shunt and a bench load)	Proposed, awaiting Amish. No recommendation was made
14	TRL 3 engineering proposals from CGD-CAL-001: protection thresholds (SCD 100 mV with 15 μ s delay, OCD1 24 mV with 320 ms, OCD2 40 mV with 20 ms, OCC 12 mV), a top-side cooled MOSFET package, a DPAK-class precharge switch, one-point current calibration at build	Engineering proposals, awaiting Amish's confirmation
15	Responses to the requirements CGD-CAL-001 finds not met (R6 board loss, R8 state of charge on small packs, R14 mass, R16 cost): options are listed in CGD-CAL-001 section 12	Proposed, awaiting Amish. No choice has been made

■ Consequences

- R4 is closed on paper by the secondary protector, but it depends on finding an SCP fuse rated for 40 A at 60 V DC or more; until then R4 is at risk.
- The SCP fuse adds about 0.6 W at 40 A on the board, which pushes the board loss to about 4.4 W against the 4 W of R6.
- Parts cost rises to \$134. R16 is not met against the \$120 in `project.yaml` and would be met against the recommended \$140.
- The SwapCell message set fixes the bus at 250 kbit/s. MotionCore, which plans to read CellGuard faults over CAN, must run its bus at the same rate.
- If CellGuard serves as SwapCell's BMS, it must also carry SwapCell's INTERLOCK wake input (about 30 μ A in sleep); CGD-CAL-001 shows sleep stays within SwapCell's 100 μ A limit.
- The BQ77216 overvoltage threshold (3.55 to 5.1 V) is chosen by orderable part option, so an LFP build and an NMC build need different part variants.